Band-1 Hadamard order 14012 – Classical Paley-I (q=14011); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14012 Paley-I PASS; Paley-II FAIL/SKIP (q=7005=3*5*467 NOT prime; ===1 mod 4 but composite); Kronecker FAIL/SKIP (4*3503; 3503===3 mod 4 NOT Hadamard-admissible; 3503=31*113 (odd composite)); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-08

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-07_Band1_Hadamard_14000_PaleyI`; `2026-10-07_Band1_Hadamard_13968_PaleyI`; `2026-10-07_Band1_Hadamard_13964_PaleyI`; `2026-10-07_Band1_Hadamard_13932_PaleyI`; `2026-10-07_Band1_Hadamard_13908_PaleyI`; `2026-10-07_Band1_Hadamard_13904_PaleyI`; `2026-10-07_Band1_Hadamard_13884_PaleyI`; `2026-10-07_Band1_Hadamard_13880_PaleyI`; `2026-10-07_Band1_Hadamard_13860_PaleyI`; `2026-10-07_Band1_Hadamard_13832_PaleyI`; `2026-10-07_Band1_Hadamard_13808_PaleyI`; `2026-10-07_Band1_Hadamard_13800_PaleyI`; `2026-10-07_Band1_Hadamard_13764_PaleyI`; `2026-10-07_Band1_Hadamard_13760_PaleyI`; `2026-10-07_Band1_Hadamard_13752_PaleyI`; `2026-10-07_Band1_Hadamard_13724_PaleyI`; `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`; `2026-10-06_Band1_Hadamard_13680_PaleyI`; `2026-10-06_Band1_Hadamard_13628_PaleyI`; `2026-10-06_Band1_Hadamard_13620_PaleyI`; `2026-10-06_Band1_Hadamard_13592_PaleyI`; `2026-10-06_Band1_Hadamard_13568_PaleyI`; `2026-10-06_Band1_Hadamard_13524_PaleyI`; `2026-10-06_Band1_Hadamard_13500_PaleyI`; `2026-10-06_Band1_Hadamard_13488_PaleyI`; `2026-10-06_Band1_Hadamard_13464_PaleyI`; `2026-10-06_Band1_Hadamard_13452_PaleyI`; `2026-10-06_Band1_Hadamard_13412_PaleyI`; `2026-10-06_Band1_Hadamard_13400_PaleyI`; `2026-10-06_Band1_Hadamard_13368_PaleyI`; `2026-10-06_Band1_Hadamard_13340_PaleyI`; `2026-10-06_Band1_Hadamard_13332_PaleyI`; `2026-10-06_Band1_Hadamard_13328_PaleyI`; `2026-10-06_Band1_Hadamard_13292_PaleyI`; `2026-10-06_Band1_Hadamard_13268_PaleyI`; `2026-10-06_Band1_Hadamard_13260_PaleyI`; `2026-10-06_Band1_Hadamard_13220_PaleyI`; `2026-10-06_Band1_Hadamard_13188_PaleyI`; `2026-10-06_Band1_Hadamard_13184_PaleyI`; `2026-10-06_Band1_Hadamard_13172_PaleyI`; `2026-10-05_Band1_Hadamard_13164_PaleyI`; `2026-10-05_Band1_Hadamard_13160_PaleyI`; `2026-10-05_Band1_Hadamard_13152_PaleyI`; `2026-10-05_Band1_Hadamard_13148_PaleyI`; `2026-10-05_Band1_Hadamard_13128_PaleyI`; `2026-10-05_Band1_Hadamard_13104_PaleyI`;

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2026-10-04_Band1_Hadamard_12792_PaleyI; 2026-10-04_Band1_Hadamard_12764_PaleyI;
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2026-09-30_Band1_Hadamard_11444_PaleyI; 2026-09-30_Band1_Hadamard_11424_PaleyI;
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2026-09-30_Band1_Hadamard_11384_PaleyI; 2026-09-30_Band1_Hadamard_11352_PaleyI;
2026-09-30_Band1_Hadamard_11312_PaleyI; 2026-09-30_Band1_Hadamard_11300_PaleyI;
2026-09-30_Band1_Hadamard_11288_PaleyI; 2026-09-30_Band1_Hadamard_11280_PaleyI;
2026-09-30_Band1_Hadamard_11252_PaleyI; 2026-09-29_Band1_Hadamard_11244_PaleyI;
2026-09-29_Band1_Hadamard_11240_PaleyI; 2026-09-29_Band1_Hadamard_11172_PaleyI;
2026-09-29_Band1_Hadamard_11160_PaleyI; 2026-09-29_Band1_Hadamard_11132_PaleyI;
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2026-09-29_Band1_Hadamard_11028_PaleyI; 2026-09-29_Band1_Hadamard_11004_PaleyI;
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2026-09-29_Band1_Hadamard_10848_PaleyI; 2026-09-29_Band1_Hadamard_10832_PaleyI;
2026-09-29_Band1_Hadamard_10800_PaleyI; 2026-09-29_Band1_Hadamard_10772_PaleyI;
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2026-09-28_Band1_Hadamard_10688_PaleyI; 2026-09-28_Band1_Hadamard_10668_PaleyI;
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2026-09-28_Band1_Hadamard_10608_PaleyI; 2026-09-28_Band1_Hadamard_10568_PaleyI;
2026-09-28_Band1_Hadamard_10560_PaleyI; 2026-09-28_Band1_Hadamard_10532_PaleyI;
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2026-09-24_Band1_Hadamard_10392_PaleyI; 2026-09-24_Band1_Hadamard_10344_PaleyI;
2026-09-24_Band1_Hadamard_10332_PaleyI; 2026-09-24_Band1_Hadamard_10304_PaleyI;
2026-09-24_Band1_Hadamard_10272_PaleyI; 2026-09-23_Band1_Hadamard_10268_PaleyI;
2026-09-23_Band1_Hadamard_10224_PaleyI; 2026-09-23_Band1_Hadamard_10212_PaleyI;
2026-09-23_Band1_Hadamard_10164_PaleyI; 2026-09-23_Band1_Hadamard_10160_PaleyI;
2026-09-23_Band1_Hadamard_10152_PaleyI; 2026-09-23_Band1_Hadamard_10140_PaleyI;
2026-09-23_Band1_Hadamard_10112_PaleyI; 2026-09-23_Band1_Hadamard_10104_PaleyI;
2026-09-23_Band1_Hadamard_10100_PaleyI; 2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_14012_paleyI/`.

Cutoff. 2026-10-08T05:32:03Z (~1:32AM ET).

Honesty stamp (non-negotiable).

This memo claims **finite construction of order 14012 only (verified max|HHˆT-nI|=0), via classical Paley-I q=14011 (prime ===3 mod 4). 14012 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14012: Paley-I **PASS** ($q = 14011$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=7005=3*5*467 NOT prime; ===1 mod 4 but composite); Kronecker **FAIL/SKIP** (note $14012 = 4 \cdot 3503$;

$3503 \equiv 3 \pmod 4$ NOT Hadamard-admissible; $3503 = 31 \cdot 113$ (odd composite)). **14012** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14000+13968+13964+13932+13908+13904+13884+13880+13860+13832+13808+13800+13764+137** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **14012**.

- H_{14012} = Paley-I ($q = 14011$) – classical SPK
- Paley-II for 14012: **FAIL/SKIP** ($q = 7005 = 3 \cdot 5 \cdot 467$ NOT prime; $\equiv 1 \pmod 4$ but composite)
- Kronecker for 14012: **FAIL/SKIP** ($4 \cdot 3503$; $3503 \equiv 3 \pmod 4$ NOT Hadamard-admissible; $3503 = 31 \cdot 113$ (odd composite))
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14012.py`

Results. overall_pass=True; units 636/636; row blocks 110/110 (block size 128); wall 2441.0207s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-08T05:32:03Z (~1:32AM ET)

Arithmetic

- Target $n = 14012 = q + 1$ with $q = 14011$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 7005 = 3 \cdot 5 \cdot 467$ NOT prime; $\equiv 1 \pmod 4$ but composite — FAIL/SKIP.
- Kronecker: $14012 = 4 \cdot 3503$; $3503 \equiv 3 \pmod 4$ NOT Hadamard-admissible; $3503 = 31 \cdot 113$ (odd composite) — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14012 Paley-I	Paley-I q=14011	0	2403.7887

Object	Method	max|HHˆT-nI|	Wall (s)
H_14012 Paley-II	FAIL/SKIP (q=7005=3*5*467 NOT prime; ===1 mod 4 but composite)	–	–
H_14012 Kronecker	FAIL/SKIP (4*3503; 3503===3 mod 4 NOT Hadamard-admissible; 3503=31*113 (odd composite))	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; …; **NOT** 8948/8644/8516/…) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=14011)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^271317 *NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^259^2 *===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^23*1163 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3^25311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2^2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*19*67 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2^23*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (FIRST n===0 mod 4 after 14012 with any direct classical Paley route; 14016/14020/14024 have none; q2=7013 prime ===1 mod 4; Paley-I q=14027=13^283 *NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2^237167 *===12 mod 16; needs a Paley-II wave (template hadamard_9124_paleyII)) | Kronecker-only (no direct Paley route; note-only, needs a Kronecker wave, NOT on box): 14032=4*3508 (3508=2^2877 ===0 mod 4; H_3508 via Paley-II q=1753 prime ===1 mod 4) and 14048=4*3512 (3512=2^3439 ===0 mod 4; H_3512 via Paley-I q=3511 prime ===3 mod 4) | next Paley-I DIRECT: q=14051->n=14052 (prime ===3 mod 4; 14029/14033 primes ===1 mod 4 skipped; no prime ===3 mod 4 in 14012..14050; Paley-II q=7025=5^2*281 NOT prime (===1 mod 4 but composite) -> FAIL/SKIP; Kronecker 14052=4*3513; 3513=3*1171 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3513); 14052=2^23*1171 ===4 mod 16 = odd-m shape with m===1 mod 4 (13924/13956/14004-like residues)) (or gap 2052 / Track B)